

Computing and Software
SFWRENG 2C03
Data Structures and Algorithms
Winter 2026



ENGINEERING

Class Format

This is an In Person class (3 hours lecture, 1 hour tutorial). Attendance and in-person participation in lecture and tutorial activities is strongly recommended.

Please note that there are no tutorials on Monday, January 5.

Course Dates: 01/05/2026 - 04/07/2026

Units: 3.00

Course Delivery Mode: In Person

Course Description: Basic data structures: stacks, queues, hash tables, and binary trees; searching and sorting; graph representations and algorithms, including minimum spanning trees, traversals, shortest paths; introduction to algorithmic design strategies; correctness and performance analysis. Implementation of data structures and algorithms. Three lectures, one tutorial (one hour); second term Prerequisite(s): SFWRENG 2DM3; registration in any Software Engineering program Antirequisite(s): COMPSCI 2C03, COMPENG 3SM4

Important Links

- [Mosaic](#)
- [Avenue to Learn](#)
- [Student Accessibility Services - Accommodations](#)
- [McMaster University Library](#)
- [eReserves](#)

- [Computing and Software Drop-in Help](#)
- [Equity and Inclusion](#)
- [2SLGBTQIA+ Resources](#)
- [Black Lives Matter](#)
- [Indigenous Student Services](#)
- [MES Clubs](#)
- [Counselling and Well Being](#)
- [TalkSpot](#)
- [Student Success Centre](#)
- [Academic Advising](#)
- [Student Technical Support](#)
- [IntelliJ IDEA Accessibility](#)
- [Google Chrome Accessibility](#)

Course Learning Outcomes

- 1. Describe the properties, common implementations, and use-cases of well-known data structures (stacks, queues, trees, hash tables, graphs, etc.)
- 2. Identify the time and space trade-offs in the implementations of data structures and algorithms.
- 3. Verify the properties of algorithms and data structures by conducting experiments in a high-level programming language.

- 5. Design and implement algorithms to solve computational problems involving stack, queue, list, tree, map, and graph Abstract Data Types (ADTs).
- 4. Determine basic properties of various data structures, algorithms, and complexity classes through formal analysis.
- 6. Justify the selection of data structures and algorithms to solve a variety of computing problems.

Graduate Attributes

The Canadian Engineering Accreditation Board (CEAB) is a division of Engineers Canada and is responsible for accrediting undergraduate engineering programs across Canada. Accreditation by the CEAB ensures that the engineering programs meet a national standard of quality and cover essential educational requirements. Graduate Attributes are a set of qualities and skills that the CEAB expects engineering graduates to possess. These attributes are a benchmark for the learning outcomes of accredited engineering programs. This section lists the Graduate Attribute Indicators associated with the Learning Outcomes in this course.

- Graduate Attribute Indicator 1.3: Competence in Engineering Fundamentals
- Graduate Attribute Indicator 1.4: Competence in Specialized Engineering knowledge
- Graduate Attribute Indicator 2.1: Identifies and states reasonable assumptions and suitable engineering fundamentals, before proposing a solution path to a problem
- Graduate Attribute Indicator 2.2: Proposes problem solutions supported by substantiated reasoning, recognizing the limitations of the solutions
- Graduate Attribute Indicator 3.1: Selects appropriately from relevant knowledge base to plan appropriate data collection methods and analysis strategies
- Graduate Attribute Indicator 7.1: Demonstrates comprehension of technical and non-technical instructions and questions
- Graduate Attribute Indicator 7.2: Composes an effective written document for the intended audience

Course Schedule

Wk	Dates	Content	Zybook	Print	Evaluation
Preliminaries					
1	Jan 5, 7, 9	Introduction, Object-Oriented Programming in Java	1.2, 2	1.2, 2	Asst 1 (5%)
2	12, 14, 16	Arrays and Linked Lists	3.1–3.3, 3.6, 3.8–3.10	3.1.1–3.1.3, 3.2, 3.4–3.6	Asst 2 (5%)
Analysis and Linear ADTs					
3	19, 21, 23	Algorithm Analysis	4, 5.1, 5.2	4, 5.1, 5.2	Asst 3 (5%)
4	26, 28, 30	Stacks, Queues, and Lists	6.1, 6.2 (skip circular queue), 7.1–7.3, 7.5	6.1, 6.2 – 6.2.4, 7.1–7.4	Asst 4 (5%)
Trees					
5	Feb 2, 4, 6	Tree ADT, tree traversal Test 1 (weeks 1 to 4)	8.1–8.3 (skip linked structure for general trees), 8.5	8.1, 8.2, 8.3.1, 8.4.1–8.4.4, 8.4.6	*** Test 1, Feb 6 (10%), LRW B1007
6	9, 11, 13	Priority Queue ADT, Heaps, Heapsort	9.1–9.4, 8.4	9.1–9.5, 8.3.2	Asst 5 (5%)
BREAK WEEK (Feb 16 to 22)					
Searching and Sorting					
7	23, 25, 27	Map ADTs, Hash Tables	10.1–10.3	10.1–10.3	Asst 6 (5%)
8	Mar 2, 4, 6	Binary Search Trees, AVL Trees	11.1–11.3	11.1–11.3	Asst 7 (5%)
9	9, 11, 13	Merge Sort, Quick Sort, Radix Sort Test 2 (Weeks 5 to 8)	12.1–12.4 (in 12.1, skip <i>recurrence equations</i> onwards)	12.1–12.4(skip 12.1.4, 12.1.5)	*** Test 2, Mar 13 (10%), LRW B1007
Graphs					

10	16, 18, 20	Graph ADTs, graph traversal	14.1–14.4	14.1–14.3	Asst 8 (5%)
11	23, 25, 27	Topological Sort, Weighted Graphs, Dijkstra's Algorithm	14.6, 14.7	14.5, 14.6	Asst 9 (5%)
12	30, Apr 1	Minimum Spanning Trees (no class April 3)	14.8, 14.9	14.7	
13	6	Review			
EXAM PERIOD (Final Exam)					

Required Materials and Texts

Please sign in with your MacID [here](#) to view your booklist

Data Structures and Algorithms in Java

Authors: Michael T. Goorich, Roberto Tamassia, and Michael H. Goldwasser

Publisher: Wiley

Publication Date: 2014 (print), 2024 (zybooks)

Edition: Sixth Edition or Zybooks Edition

E-book starts at \$27, Zybooks edition is \$54.50

Course Evaluation

There are two possible marking schemes for the course. We will use whichever one results in a higher grade for you.

Marking Scheme 1

- **Tutorials:** 5% (10 of 12)
- **Assignments:** 45% (9 equally weighted)
- **Midterm and Final Exams:** 10%, 10%, 30%

Marking Scheme 2

- **Tutorials:** 5% (10 of 12)
- **Assignments:** 45% (9 equally weighted)
- **Midterm and Final Exams:** 0%, 0%, 50%

Course Evaluation Details

Tutorials

Most tutorials will focus on exercises concerning the formal properties of data structures and algorithms. Attendance and participation in 10 of the 12 tutorials is required.

Assignments

Each assignment will consist of Java coding exercises and written analysis related to the most recently covered content. This will involve expanding or refining an existing Java codebase for data structures, designing and analyzing algorithms, implementing algorithms in Java, code experiments in Java, mathematical proofs and derivations, and written analyses of data structures, algorithms and their implementations.

You are encouraged to discuss the course material with your classmates, but your assignments must be your own individual effort. You may consult other sources, but all such sources must be documented. Failure to do so may result in academic dishonesty charges.

Exams

All exams will be closed-book, paper-and-pencil tests. There will be two midterms written during class in weeks 5 and 9 (check the schedule above for the date). The midterms will cover the most recent material from the lectures and tutorials. The final exam will be scheduled after the end of classes. The final exam will be cumulative.

Late Assignments

Assignments are due on the day listed on the assignment handout (i.e., up to 11:59:59 pm that day). After that, they are late. Late days are tracked by calendar date. For example, if the due date is November 1st and you hand the assignment in some time on November 2nd, it is one day late.

You can hand in late assignments with no penalty, but you have a limit of 12 late days for all assignments combined, and you can use a maximum of 3 late days on each assignment without requiring an MSAF. If you go over either limit, late assignments will no longer be accepted. Please note that unless your instructor states otherwise, the final day to hand in any work for this course, regardless of how many late days you have remaining, will be the last day of classes.

Absences, Missed Work, Illness

Tutorials

You can skip two tutorials with no penalty. After that, each tutorial is worth 0.5% of your final grade. If you MSAF a tutorial, you will be marked as present for that tutorial.

Assignments

Your 12 late days do not include special or unforeseen circumstances (illnesses, family emergencies, etc.). If you are delayed in completing an assignment due to special or unforeseen circumstances, let your professor know as soon as you can so that you can negotiate a new due date. Keep in mind that you may be required to produce supporting documentation. If you MSAF an assignment, we can either shift the weight to the final exam or give you some more time to complete the assignment.

Midterm Exams

If you miss the midterm exam for a legitimate reason (MSAF), the weight of that exam will be moved to the final exam.

Generative AI: Use Prohibited

Students are not permitted to use generative AI in this course. In alignment with [McMaster academic integrity policy](#), it "shall be an offence knowingly to ... submit academic work for

assessment that was purchased or acquired from another source". This includes work created by generative AI tools. Also state in the policy is the following, "Contract Cheating is the act of "outsourcing of student work to third parties" (Lancaster & Clarke, 2016, p. 639) with or without payment." Using Generative AI tools is a form of contract cheating. Charges of academic dishonesty will be brought forward to the Office of Academic Integrity.

APPROVED ADVISORY STATEMENTS

Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the [Academic Integrity Policy](#).

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

Authenticity / Plagiarism

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online

learning platform (e.g. Avenue to Learn, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. **All submitted work is subject to normal verification that standards of academic integrity have been upheld** (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

Courses with an On-line Element

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and

working communities. These expectations are described in the [Code of Student Rights & Responsibilities](#) (the “Code”). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online.**

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students’ access to these platforms.

Academic Accommodation of Students with Disabilities

Students with disabilities who require academic accommodation must contact [Student Accessibility Services](#) (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University’s [Academic Accommodation of Students with Disabilities](#) policy.

Requests for Relief for Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).

Academic Accommodation for Religious, Indigenous, or Spiritual Observances (RISO)

Students requiring academic accommodation based on religious, Indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should submit their request to their Faculty Office ***normally within 10 working days*** of the beginning of term in which they anticipate a need for accommodation or to the Registrar’s

Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Copyright and Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.